Unit 1 Quiz: Displaying One Variable KEY

Name _____

Show your work.

1.1 - Frequency Tables, Dot Plots, and Histograms

1. **I make and read a frequency table and a dot plot.**

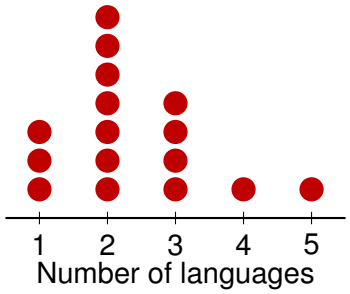
   16 students said how many languages they speak.

   2, 1, 3, 2, 4, 2, 1, 3, 2, 3, 2, 1, 5, 2, 3, 2

   (a) Make a frequency table.

Languages	Tally	Frequency
1	III	3
2	IIIIIII	7
3	IIII	4
4	I	1
5	I	1

   (b) Make a dot plot.

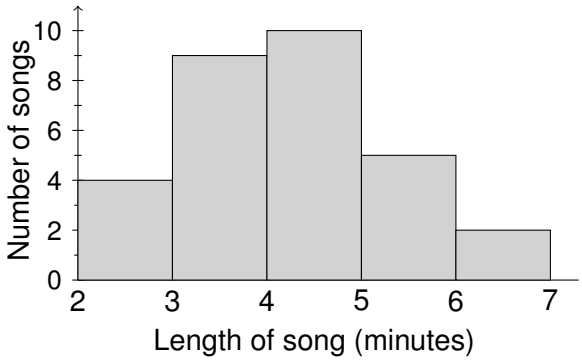


   (c) How many students speak 3 or more languages?

   4 + 1 + 1 = 6 students

2. **I read a histogram.**

   This histogram shows the lengths of 30 songs on a playlist.



   (a) How many songs are from 3 to 4 minutes long?

   9 songs

   (b) How many songs are 5 minutes long or longer?

   5 + 2 = 7 songs

   (c) One song is exactly 4 minutes long. Which bar is it in?

   The 4 to 5 bar. A value on the line goes in the bar on the right.

   (d) Can you tell from the graph how long the longest song is? Explain.

   No. It is from 6 to 7 minutes long. A histogram hides the exact values.

1.2 - Mean, Median, and Shape

3. **I find the mean and the median of a data set.**

A basketball player scored these points in 6 games.

$$14,\ 9,\ 21,\ 12,\ 9,\ 19$$

(a) Find Σx.

$\Sigma x = 14 + 9 + 21 + 12 + 9 + 19 = 84$

(b) Find the mean, $\bar{x}$.

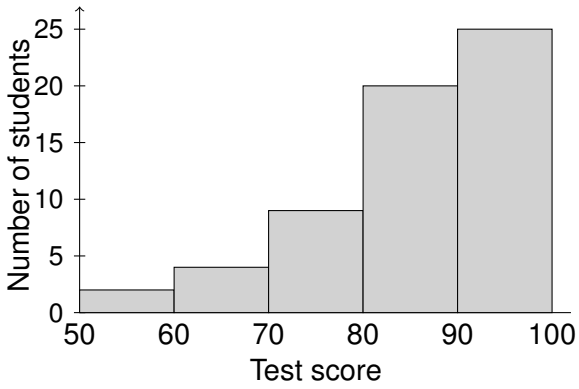
$n = 6, \quad \bar{x} = \dfrac{\Sigma x}{n} = \dfrac{84}{6} = 14$ points

(c) Find the median.

In order: 9, 9, $\boxed{12}$, $\boxed{14}$, 19, 21. Median = $(12 + 14) \div 2 = 13$ points

4. **I tell the shape of a graph, and I choose the best measure of center.**

This histogram shows the scores of 60 students on a test.



(a) Circle the shape of the data.

(skewed left) symmetric skewed right

(b) Which is greater, the mean or the median? Explain.

The median. The tail is on the left. The tail pulls the mean down, so the mean is less than the median.

(c) Which statement is true? Circle the letter.

A The mean best describes a typical score, because the data are skewed left.
(B) The median best describes a typical score, because the data are skewed left.
C The mean best describes a typical score, because the data are symmetric.
D The median best describes a typical score, because the data are symmetric.

1.3 - Quartiles, IQR, and Box Plots

5. **I find the five-number summary and the IQR.**

11 students counted how many push-ups they did in one minute.

18, 25, 12, 30, 22, 15, 27, 20, 35, 16, 24

(a) Find the five-number summary.

In order: 12, 15, 16, 18, 20, 22, 24, 25, 27, 30, 35. Median 22.
Lower half 12, 15, 16, 18, 20: Q1 = 16. Upper half 24, 25, 27, 30, 35: Q3 = 27.

Min 12 Q1 16 Median 22 Q3 27 Max 35

(b) Find the IQR.

IQR = 27 − 16 = 11 push-ups

6. **I make a box plot.**

12 students counted how many songs they listened to today.

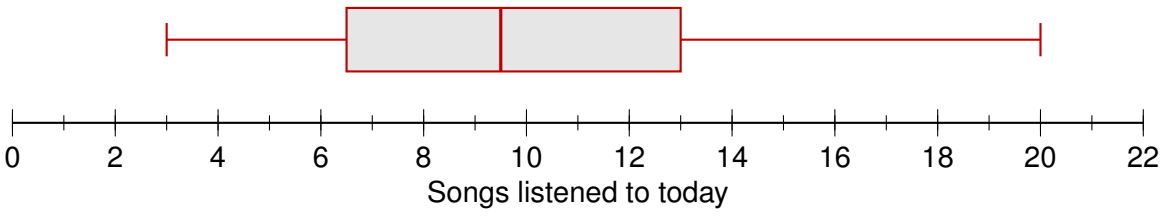
8, 3, 15, 10, 6, 12, 20, 9, 5, 14, 11, 7

(a) Find the five-number summary.

In order: 3, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 20. Median = (9 + 10) ÷ 2 = 9.5
Lower half 3, 5, 6, 7, 8, 9: Q1 = (6 + 7) ÷ 2 = 6.5
Upper half 10, 11, 12, 14, 15, 20: Q3 = (12 + 14) ÷ 2 = 13

Min 3 Q1 6.5 Median 9.5 Q3 13 Max 20

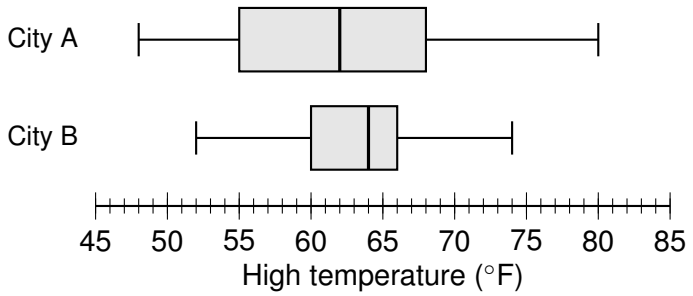
(b) Make a box plot.



1.4 - Reading Box Plots and Outliers

7. **I read and compare box plots.**

These box plots show the high temperature each day in April in two cities.



(a) Which city has the higher median?

City B (64°F is more than 62°F)

(b) Find the IQR of each city.

City A: $68 - 55 = 13$°F
City B: $66 - 60 = 6$°F

(c) Which city has more consistent temperatures? Explain.

City B. Its IQR (6) and its range (22) are both smaller than City A's (13 and 32).

(d) About what percent of the days in City A had a high temperature above 55°F?

About 75%. 55 is Q1 for City A.

8. **I use the 1.5 × IQR rule to find outliers.**

9 customers recorded how many minutes they waited in line at a store.

13, 3, 18, 10, 34, 15, 12, 20, 16

(a) Find Q1, Q3, and the IQR.

In order: 3, 10, 12, 13, $\boxed{15}$, 16, 18, 20, 34. Median 15. Leave it out.
Lower half 3, 10, 12, 13: Q1 = $(10 + 12) \div 2 = 11$
Upper half 16, 18, 20, 34: Q3 = $(18 + 20) \div 2 = 19$ IQR = $19 - 11 = 8$

(b) Find the lower fence and the upper fence.

$1.5 \times 8 = 12$
Lower fence = $11 - 12 = -1$ Upper fence = $19 + 12 = 31$

(c) Which values are outliers? Explain.

34 is an outlier, because it is above 31. 3 is **not** an outlier, because it is not below -1.

(d) On a box plot of this data, where does the right whisker end? Why?

At 20. The whisker stops at the largest value that is not an outlier.